

Literature Status: Eigenvalue Multiplicity in General Left-Definite Theory

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Status: literature implied answer. Conjecture 3 of arXiv:2012.01014 follows immediately from the unitary equivalence in Littlejohn–Wellman, Theorem 6.2(iii). This is an identification of an earlier theorem, not a new result of the run.

1 Original conjecture

D. Frymark and C. Liaw, *Perspectives on General Left-Definite Theory*, arXiv:2012.01014, state the following as Conjecture 3 on arXiv PDF page 10 [1].

Source Conjecture 3. *Let A be a semibounded self-adjoint operator in a Hilbert space H . If λ is an eigenvalue of A of multiplicity m , then λ is an eigenvalue of the r -th left-definite operator A_r in H_r , again of multiplicity m .*

In the native normalization of general left-definite theory, $A \geq kI$ for some $k > 0$, $H_r = \text{dom}(A^{r/2})$ with

$$\langle x, y \rangle_r = \langle A^{r/2}x, A^{r/2}y \rangle_H.$$

A merely semibounded operator is handled by the standard positive scalar shift used in the source.

2 Earlier theorem and the implication

L. L. Littlejohn and R. Wellman, *On the Spectra of Left-Definite Operators*, prove in Theorem 6.2(iii), PDF page 11, that A and every A_r are unitarily equivalent [2]. More precisely,

Earlier theorem. *The map $U_r = A^{r/2} : H_r \rightarrow H$ is unitary, maps $\text{dom}(A_r)$ bijectively onto $\text{dom}(A)$, and satisfies*

$$U_r A_r U_r^{-1} = A.$$

For any $\lambda \in \mathbb{R}$, the intertwining identity gives

$$U_r(\ker(A_r - \lambda I)) = \ker(A - \lambda I).$$

Indeed, $A_r x = \lambda x$ if and only if $A(U_r x) = U_r(A_r x) = \lambda U_r x$, and the converse follows using U_r^{-1} . Thus the two eigenspaces are linearly isomorphic, so their dimensions are equal. Since eigenvalue multiplicity for a self-adjoint operator is the dimension of its eigenspace, Conjecture 3 follows in full.

For an operator only bounded below, choose c so $B = A + cI \geq kI > 0$ and apply the theorem to B . The same unitary intertwines the correspondingly shifted left-definite realization of A with A ; subtracting cI restores the eigenvalue λ and does not change eigenspace dimensions.

Provenance and scope

The supporting paper predates the source conjecture, so it does not explicitly claim to answer it. The source cites this paper for spectral stability but does not state the above multiplicity consequence. This packet therefore records an agent-identified direct implication, not a new proof and not an author-explicit later answer.

A bounded search through 9 August 2026 checked the run indexes, exact conjecture wording, the source's citations, and arXiv/web searches for left-definite eigenvalue multiplicity and unitary equivalence. The decisive source was Littlejohn–Wellman Theorem 6.2. The separate informal question about interlacing for two different self-adjoint extensions is not settled by this identification.

References

- [1] D. Frymark and C. Liaw, *Perspectives on General Left-Definite Theory*, arXiv:2012.01014 (2020).
- [2] L. L. Littlejohn and R. Wellman, *On the Spectra of Left-Definite Operators*, Complex Anal. Oper. Theory **7** (2013), 437–455. doi:10.1007/s11785-011-0178-6.